TJM²

FEBRUARY 2027

TJHSST, Alexandria · date soon

A one-day math modeling competition for middle and high school students

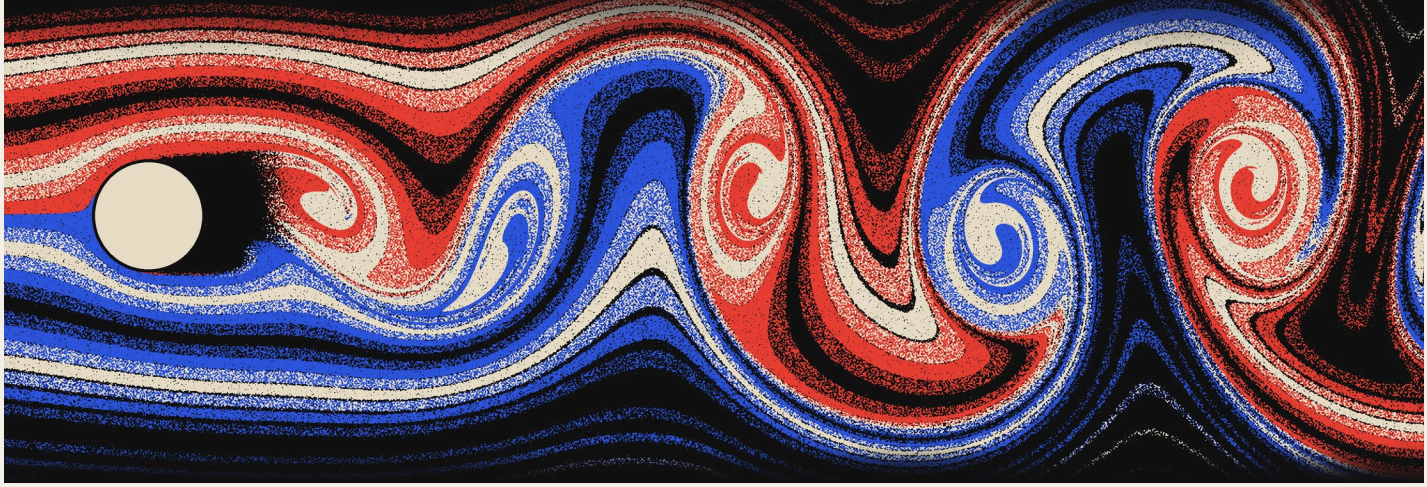


Fig. 1 **Will Northern Virginia's data centers outgrow the power grid?**
SAMPLE The real problem is revealed at 9 a.m. on competition day.

1 day	**6–12**	**3–5**	**$2,000+**	**Free**
8 a.m. to 7 p.m.	grades, two divisions	per team, or come solo	in prizes	food included

HOW IT WORKS

1. **Get the problem.**
   Open-ended, built on real data, with no single right answer.
2. **Build a model with your team.**
   Math, code, spreadsheets: whatever works. Papers are due mid-afternoon.
3. **Defend it.**
   Finalists present to judges; awards and raffles close the night.

AI policy. Use AI to find models and write code. Judges score how well you can justify every decision.

WHY COME

- **$2,000+ in prizes**, swag for every team that turns in a paper, and raffles.
- **A day out with friends**, with free lunch, dinner and snacks.
- **No experience needed.** We teach the basics that morning and answer questions on our Discord.
- **TJ students:** top performers are considered for our M3 Challenge team.

SCHEDULE

8:00	Check-in, breakfast
8:45	Opening ceremony
9:00	**Problem released**
12:00	Lunch
3:00	**Papers due**
3:15	Games while judges read
4:45	Finalist presentations
6:15	Dinner
6:30	**Awards and raffle**

Tentative; details by email.

SAMPLE PROBLEM

Not this year's

Northern Virginia is the world's largest data-center market, and Dominion Energy projects its Virginia electricity load could roughly double within 15 years. **(1)** Model a data center's power draw across a year. **(2)** Forecast the region's demand 5–15 years out. Will it outgrow supply? Quantify your uncertainty. **(3)** Recommend where the next ones should go.

Data provided: EIA Form 861 · PJM load forecasts · Dominion's Integrated Resource Plan · JLARC data-center study (2024) · Loudoun County GeoHub. See a winning paper: TJ's 2023 M3 Challenge champion.

Sign up to hear more

tjmmc.org/tjm2

Non-binding. Coming solo? Sign up anyway and we'll match you with a team.
Questions: officers@tjmmc.org

Hosted by the **TJ Math Modeling Club**
M3 Challenge champions, 2023 · runners-up, 2024